电子电路与系统基础期末 Cheat Sheet

一、正弦稳态分析：（正弦信号 Sinusoidal Signal）

Thm: 在正弦信号激励下，线性时不变电路中所有电压电流变化与激励源相同频率正弦信号，只有幅度相位发生变化。（正弦稳态响应 Sinusoidal Steady-State）.

def 相量（Phasor）若 $p(t)=A\cos(\omega t+\phi)$, $P=A\angle\phi = Ae^{j\phi}$.（同一频率才可同时用相量表示）

Thm: 相量表示下电路定律：形式不变（注意：直流源在AC分析中置零）交流源同频 ↗

Thm: 相量微积分：$\frac{d}{dt}\to j\omega$（相位提前90°）. $\int \to \frac{1}{j\omega}$. 相位落后90°

def: 复阻抗参数. $Z = R + jX$（单位：欧姆）

（impedance）阻抗　电阻（resistance）电抗（reactance）　$\boxed{Y=Z^{-1}}$

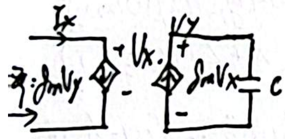
$Y = G + jB$.（单位西门子）

（Admittance）导纳　电导（conductance）电纳（Susceptance）

def: 感性（Inductive）: $\mathrm{Im}\,Z > 0$. $\mathrm{Im}(Y)<0$ 电压超前电流

容性（capacitive）$\mathrm{Im}\,Z < 0$　$\mathrm{Im}(Y) > 0$. 电压落后电流

note: 感/容性未必有这些元件！例如回转器电路

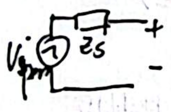


$\dot V_y = g_m \dot V_x \cdot \frac{1}{j\omega C}$ ⇒ $Z_{eq} = \frac{\dot V_x}{\dot I_x}$

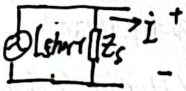
$\dot I_x = g_m \dot V_y = \frac{g_m^2}{j\omega C}\cdot\dot V_x$ $= \frac{j\omega C}{g_m^2}$ 感性！

Thm: Thevinin/Norton 等效：

二端口线性时不变电路可等效为一个含阻抗 Z_s 的交流源.



或…



诺顿等效 $Z_s = \boxed{\frac{|\dot V_{open}|}{|\dot I_{short}|}\angle V_{op}-I_s}$

def: 正弦稳态下功率计算（复功率 Complex Power）

$S = P + jQ = \dot V_{rms}\cdot \dot I_{rms}^*$（rms：有效值）.

P: 有功功率（Active Power）实际消耗或输出（即元件平均功率）

Q: 无功功率（Reactive Power）在电源/元件中来回交换功率.

$\cos\phi$: 功率因数（Power Fact, PF）　$\phi=\arg S$

Thm: 最大功率传输：共轭匹配 $Z_L = \overline{Z_s}$

note: Q ~~$\cos$~~ >0 / $\cos\phi>0$ is lagging / 感性

二、频率响应：（Frequency Response）.

↳ 正弦稳态响应的复相量随激励频率（ω）变化的函数.

记作 $H(j\omega)/H(s)$. $H(s)$ 称为传递函数（Transfer Function）

$H(j\omega) = |H(j\omega)|\cdot e^{j\angle H(j\omega)}$. $\begin{cases}\text{幅频响应 (Magnitude Response)}\\ \text{相频响应 (Phase Response)}\end{cases}$

def: 贝尔（Bel）：1贝尔 = 能量大10倍

分贝（deci-Bel）指十分之一贝尔.

$\begin{cases}\text{能量 dB} = 10\log(\frac{P_2}{P_1})\ \text{dB10}\\ \text{电压/电流 dB} = 20\log(\frac{V_2}{V_1})\ \text{dB20.}\end{cases}$

note: 常用 dB_m 表示与1mW比较，类似有 $dB_{\mu V}$. $dB_{\mu A}$，因此，相加时要同一量纲.

Thm 运算：$\begin{cases}\text{倍数相乘，贝尔相加}\\ \text{倍数相加，换算回来再算.}\end{cases}$

两等功率相加 +3dB，两个等电压相加 +6dB

相差10dB以上，相加 ~ 大的

倍数表：	0.1	0.2	0.5	1	2	3	4	5	6	7	8
dB10	−10	7.0	−3.0	0	3	4.8	6	7.0	7.8	8.5	9
dB20	−20	−14	−6	0	6	9.6	12	14	15.6	17	18

def {Bode Plot 波特图）用dB的幅频曲线与相频曲线同时在对数（角）频率坐标下进行分析

note：一些频响例子：

① 级联电路（Cascading）：一个电路的输出作为另一个输入，称为级联. 级联响应 = 各部分响应相乘 = dB相加

② 二阶RLC响应（如并联RLC）

$H(j\omega) = \frac{R}{1+jR(\omega C-\frac{1}{\omega L})}$　$\frac{R}{\sqrt{(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega})^2/4\zeta^2+1}}$

$\angle H(j\omega) = -\tan^{-1}((\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega})/2\zeta)$

def 品质因数（Q. Quantity factor）

$Q = \frac{1}{2\zeta} = R\sqrt{\frac{C}{L}}$.

3dB带宽 $|H(j\omega)| = \frac{1}{\sqrt2}|H(j\omega_0)|$ ⇒ $BW_{3dB} = \frac{\omega_0}{Q}$

def: 在频域处理信号，带有频响的电路是一个滤波器（Filter）

理想Filter / Brick Filter：精确取出频率内信号 $\begin{cases}\text{low-pass LPF}\\ \text{high-pass HPF}\\ \text{band-pass BPF}\\ \text{band-stop/notch}\end{cases}$

自然：有过渡带+阻带外有斜率，由order决定

Thm: 传函与微分方程：

$\sum_{k=0}^{n} a_k \frac{d^{(k)}x(t)}{dt^k} = \sum_{k=0}^{m} \frac{d^{(k)}y(t)}{dt^k}\cdot b_k$. 左侧特征根：极点

将右侧 $y(t)$ 分解为 $Y_i e^{s_i t}$（模态）.

对于 $y(t)$ 中 ∀ 模态，$x(t)$ 解也同模态. $x(t) = X_i e^{s_i t}$

⇒ $\frac{X_i}{Y_i} = \frac{b_m s_i^m + \dots + b_1 s_i + b_0}{a_n s_i^n + \dots + a_1 s_i + a_0} \to H_i \xrightarrow{s_i\text{遍历}} H(s)$

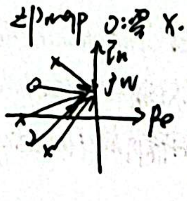
即拉普拉斯变换. $s \leftrightarrow j\omega$ 相频分析. 因 $H(s) = \frac{Z(s)}{P(s)}$.

$P(s)$ 根 p_i 为极点（Poles），决定系统自由响应模态. $s=p_i$. $x(t)=e^{p_i t}C$ [illegible]模态

$Z(s)$ 根 z_i 为零点（Zeros），决定系统屏蔽激励的模态　$s=z_i$ $H(s)=0$

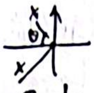
⇒ $H(s) = k\cdot\frac{\prod(s-z_i)}{\prod(s-p_i)}$ ZPK形式

ZP map　o：零　x：极



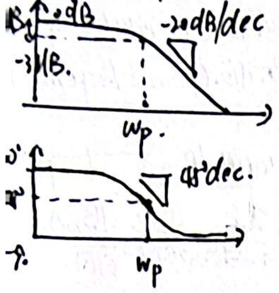
扫描全能王 创建

极点对波特图影响

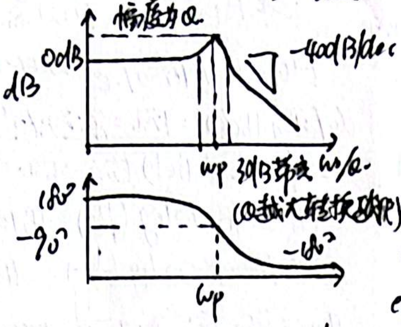


一个实极点p（一阶模型）

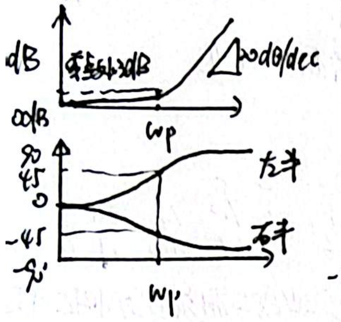
Re[p]>0 不稳定.



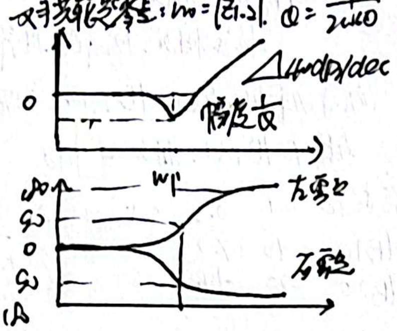
一对共轭极点 $\omega_0 = |p_{1,2}|$ $Q = \frac{1}{2\zeta} = \frac{1}{2\cos\theta}$



一个实零点 Re[z] 对应时候.



一对共轭复零点: $\omega_0 = |z_{1,2}|$. $Q = \frac{1}{2\cos\theta}$



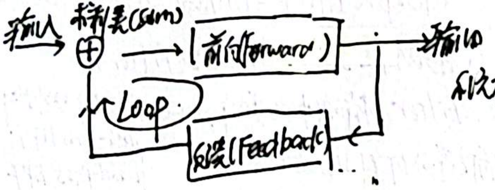
Thm: 波特图草绘/读图: { 找出转折点、下降率 → A-ω
平稳段. 下降最快. → φ-ω

Thm: 一/二阶节 (Second-Order section, SOS) 级联

将复杂传函进行分解为多个一阶/二阶传函乘积. 再以级联模块实现

三. 反馈与运算放大器

def: 单环反馈系统：前向+反馈+求和/比较模块.

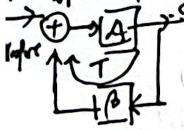


系统框图 (Block diagram).

电路中有四种反馈 (V-V, V-I, I-V, I-I)：只看量纲判断，不要被拓扑结构打乱)

def: 反馈极性: { Positive (正反馈) 反馈加强原信号使一圈后自身叠加增强
Negative (负反馈). 自身叠加抑制 → 稳定.

Yo: 静态分析



output A - 前向开环增益 (Open-Loop Gain)

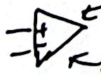
β - 反馈系数 (Feedback factor, β Factor)

Output = A×(Input ± output·β) ⇒

闭环增益 (Closed-loop Gain) $A_{fb} = \frac{A}{1\mp A\beta} = \frac{A}{1\mp T}$

反馈的特性和作用: { $A\beta = T \gg 1$ 时 $A_{fb} \approx \frac{1}{\beta}$：深度负反馈，只与β Factor
$\frac{dA_{fb}/A_{fb}}{dA/A}$ = A-Sensitivity = $\frac{1}{T}$. 开环增益误差被抑制

def: 运算放大器 Operational Amplifier



← 同相输入端 (Non-inverting Input) 输出端 假设单端运放

← 反相输入端 (Inverting Input)

Thm: 理想运放性质:

① $V_{out} = A_v(V_{in+} - V_{in-})$. 其中 $A_v \to \infty$

(Vout是对地电压，一般不画地线. 但实际需要接地/供电源)

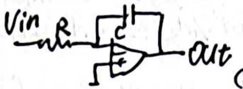
② 输入端开路 $I_{in+} = I_{in-} = 0$

③ 输出电压限幅 $V_{RAIL-} \le V_{OUT} \le V_{RAIL+}$ (正负电源轨 Supply Rail)

超出此范围，称为饱和 (saturation)

Cor: 一个运放在负反馈，饱和状态下 $V_{in+} = V_{in-}$，即虚短 (Virtual Short)

e.g. 反相积分器



① check 负反馈: Vout↑ → Vx↑ → Vout↓

② 未饱和下.

$V_x = 0$. $i_R(t) = \frac{V_{in}(t)}{R}$. $V_{out}(t) = V_x - V_C(t) = -\frac{1}{RC}\int_0^t V_{in}(t)\,dt$

四: 反馈稳定性 { External Stability 有界输入→有界输出
Internal Stability 零输入收敛至0

Thm: 波特稳定性判据: (Bode Stability Criterion).

当反馈环路的相移达到180°, 同时环路增益超过0dB, 则反馈可能不稳定. 即 $|L(j\omega_{180})| \ge 1$ 时反馈才能不稳定. 其中 ω_{180} 称为相位交越频率 (cross-over-frequency), 满足 $\angle(L(j\omega_{180})) = 180°$

note "180" 实际 180+360k 均可, 但一般看-180

def: 增益裕度 (Gain Margin) $GM(dB) = -20\log_{10}|L(j\omega_{180})|$

相位裕度 (Phase Margin) $PM(dB) = 180° + \angle(L(j\omega_{0dB}))$.

Thm: 基于极点的分析: A(s) 开环传函. β(s) 反馈系数. $L(s) = A(s)\beta(s)$

$A_{fb}(s)$ (闭环传函) $= \frac{A(s)}{1+L(s)}$

对LTI而言: 稳定 ⟺ 无实部非负极点 (右半平面极点)

闭环稳定: $A_{fb}(s)$ 没有右半平面极点

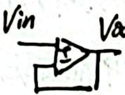
Thm: Nyquist Stability Criterion 奈奎斯特稳定性判据

$L(j\omega)$ 在复平面上的轨迹环绕 -1 的圈数为0

在Bode图上表述为:

在所有 |L| > 0dB 频段上, ∠L 从上往下与从下往上穿越 -180°+360k 次数相等

e.g. 单位增益运放.



$\frac{GBP}{s}$. $H(s) = \frac{GBP}{s}$.

$A_{fb}(s) = \frac{GBP/s}{1+GBP/s} = \frac{1}{\frac{s}{GBP}+1}$ 稳定. 等效于低通滤波器

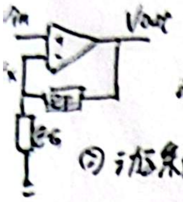
$V_{out} = V_{in}(\infty)(1 - e^{-GBP\,t})$

若反接 $A_{fb} = \frac{GBP/s}{1-GBP/s}$. 右半平面极点不稳定!

扫描全能王 创建

5.运放之应用.

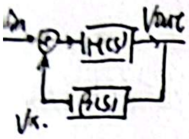
1.同相放大器：在理想运放下 $H(s)\to\infty$. $\frac{V_{out}}{V_{in}}=\frac{1}{\beta(s)}=1+\frac{Z_F}{Z_G}$


Vin
Vout
ZF
ZG

特例：$Z_G=R$. $Z_F=\frac{1}{sC}$ $\frac{V_{out}}{V_{in}}=1+\frac{1}{sRC}$

即 $V_{out}(t)=V_{in}(t)+\frac{1}{RC}\int V_{in}(\tau)d\tau+V_C(0)$.

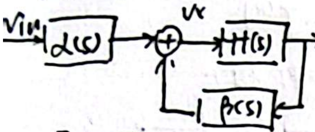
②非理想条件下. 考虑运放单极点特性. $H(s)=\frac{GBW}{s}$.


Vin
H(s)
Vout
β(s)
Vx.

$\frac{V_{out}}{V_{in}}=\frac{H(s)}{1+H(s)\beta}=\frac{1/\beta}{\frac{s}{GBW}+1}$. 极点频率GBW·β.

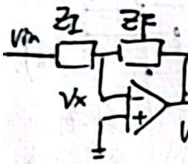
增益×带宽 =GBW

2.反相放大器


Vin
α(s)
Vx
H(s)
β(s)
Vout

$\alpha(s)=\frac{Z_F}{Z_I+Z_F}$ $\beta=\frac{Z_I}{Z_I+Z_F}$

理想条件下 $H(s)\to\infty$

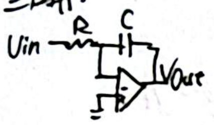

Vin
ZI
ZF
Vx
Vout

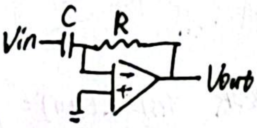
$\frac{V_{out}}{V_{in}}=-\alpha(s)\frac{1}{\beta(s)}=-\frac{Z_F}{Z_I}$ (可导出公式)

Note: 在负反馈,未饱和,理想运放下 $V_x=0$.
即虚地(Virtual ground)

特殊形式：$Z_I=0$, 则 $V_{out}=-i_{in}\cdot Z_m$ (流控电压源) ↗跨阻放大器TIA

一些应用：





反相积分器(Inverting Integrator)

$Z_I=R$ $Z_F=\frac{1}{sC}$

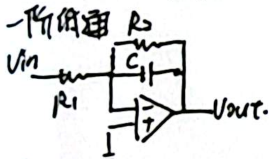
$\frac{V_{out}}{V_{in}}=-\frac{1}{sRC}$, $V_{out}(t)=V_{out}(0)-\frac{1}{RC}\int_0^t V_{in}(\tau)d\tau$

反相微分器

$Z_I=\frac{1}{sC}$. $Z_F=R$ $\frac{V_{out}}{V_{in}}=-sRC$

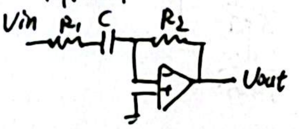
$V_{out}(t)=-RC\frac{dV_{in}(t)}{dt}$

一阶低通


Vin
R1
R2
C
Vout.

$\frac{V_{out}}{V_{in}}=-\left(\frac{R_2}{R_1}\right)\frac{1}{1+sR_2C}$

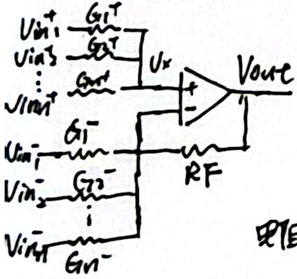
一阶高通


Vin
R1
C
R2
Vout

$\frac{V_{out}}{V_{in}}=-\left(\frac{R_2}{R_1}\right)\frac{sR_1C}{1+sR_1C}$

若 Z_1. Z_F 纯容, 则 $\frac{V_{out}}{V_{in}}=-\frac{C_1}{C_2}$. 与s无关, 构成电容电路(电容反馈反相放大器)

def: 多输入加法放大器


Vin1+
G1+
Vin2+
G2+
Vinn+
Gn+
Vx
Vout
Vin1-
G1-
Vin2-
G2-
Vinn-
Gn-
RF

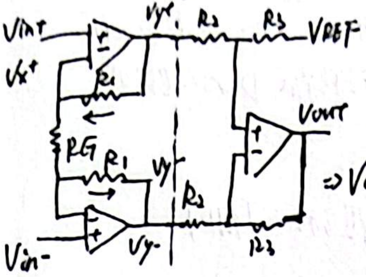
V_{out}^+ (仅正输入) $=\frac{1+R_F\sum G_i^-}{\sum G_i^+}\sum(G_i^+V_{in,i}^+)$

V_{out}^- (仅负输入) $=-R_F\sum(G_i^-V_{in,i}^-)$

$V_{out}=V_{out}^++V_{out}^-$

电阻可换为阻抗

e.g. 仪用放大器 (Instrumentation Amplifier, IA) 举例.


Vin+
Vx+
Vy+
R2
R3
VREF
R1
RG
R1
Vy-
R2
R3
Vout
Vin-
Vy-

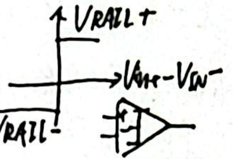
$V_{y^+}-V_{y^-}=\left(1+\frac{2R_1}{R_G}\right)(V_{in^+}-V_{in^-})$

$V_{out}=\frac{R_3}{R_2}(V_{y^+}-V_{y^-})+V_{REF}$

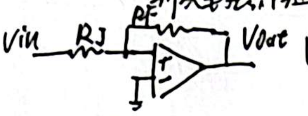
$\Rightarrow V_{out}=\left(1+\frac{2R_1}{R_G}\right)\frac{R_3}{R_2}(V_{in^+}-V_{in^-})+V_{REF}$

e.g. 运放开环使用:比较器(Comparator)

根据输入电压大小关系, 判别输出为何两个状态电平


VRAIL+
Vin+-Vin-
VRAIL-

正反馈应用: 实现迟滞特性 (hysteresis). 比较结果为低/高时 提高判决电压, 防止噪声的 False Trigger


Vin
RI
RF
Vout

$V_{TL}=-\frac{R_I}{R_F}V_{RAIL}^+$ $V_{TH}=-\frac{R_I}{R_F}V_{RAIL}^-$

(在不同状态间移动 ⇒ 较好 Latch 特性)

六.二端口网络

def: 电路网络(Circuit Network, N)是指一个电路的抽象模型.
端子(Terminal, T)是电路网络与外界相连接的节点

def 端口(Port): 一个电路中的两个端子 (TA, TB), 如果流入A的电流恒等于流出B的电流 ($i_A=-i_B$), 则AB可组成一个端口

一个端口由2个端子构成, 需指定极性.

def. 单端口网络 (Single Port Network) 只有两个端子对外连接电路 (两T一端P)

def. 多端口: 有多个端口, 行为完全由端口描述

(形为不与端口间电压存在 $F(V_1\dots V_n, I_1\dots I_n)=0$, 则是n端子网络)

note1: 计端口电路子孤立, 则一般满足

note2: 一种特殊多端口: 共地~ (common-ground multi-port network)

即N端子与地构成N端口网络

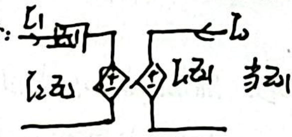
比如二端口 $\begin{cases}k_{11}V_1+k_{12}V_2+k_{13}I_1+k_{14}I_2=0\\k_{21}V_1+k_{22}V_2+k_{23}I_1+k_{24}I_2=0\end{cases}$

四.二. 6种矩阵描述

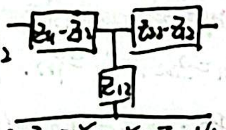
Z (阻抗参数, Impedance Parameters) $\begin{bmatrix}V_1\\V_2\end{bmatrix}=\begin{bmatrix}Z_{11}&Z_{12}\\Z_{21}&Z_{22}\end{bmatrix}\begin{bmatrix}I_1\\I_2\end{bmatrix}$.

Z_{11} 为输入阻抗 Z_{22} (输出~) Z_{12} (反向转移~) Z_{21} (正向转移~)

等效电路:

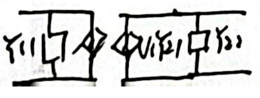

I1
Z11
I2Z12
I1Z21
Z22
I2

当 $Z_{21}=Z_{12}$

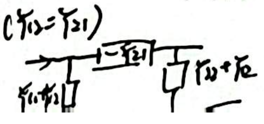

Z11-Z12
Z22-Z12
Z12

Y (导纳参数 Admittance Parametres) $\begin{bmatrix}I_1\\I_2\end{bmatrix}=\begin{bmatrix}Y_{11}&Y_{12}\\Y_{21}&Y_{22}\end{bmatrix}\begin{bmatrix}V_1\\V_2\end{bmatrix}$.

Y_{11} 为输入~ Y_{22} 为输出~ Y_{12} (反向~) Y_{21} (正向~)


Y11
V1Y21
Y22

或 ($Y_{12}=Y_{21}$)


-Y21
Y11+Y12
Y22+Y12

T传输参数 (Transmission Parameter / ABCD).

$$\begin{bmatrix}V_1\\I_1\end{bmatrix}=\begin{bmatrix}A&B\\C&D\end{bmatrix}\begin{bmatrix}V_2\\\underset{\triangle}{-I_2}\end{bmatrix}.$$

A: 开路电压比　B: 反向转移阻抗

C: 反向转移导纳　D: 短路电流比.

T' 逆传~ (Inverse~, A'B'C'D')

$$\begin{bmatrix}V_2\\I_2\end{bmatrix}=\begin{bmatrix}A'&B'\\C'&D'\end{bmatrix}\begin{bmatrix}V_1\\-I_1\end{bmatrix}$$

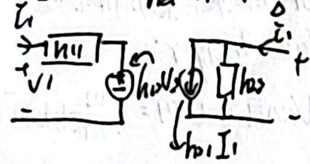
. 口值含义与T相同

h参数 (混合参数, Hybrid Parameters).

$$\begin{bmatrix}V_1\\I_2\end{bmatrix}=\begin{bmatrix}h_{11}&h_{12}\\h_{21}&h_{22}\end{bmatrix}\begin{bmatrix}I_1\\\underset{\triangle}{V_2}\end{bmatrix}$$

h_{11}: 输入阻抗　h_{12}: 反向电压增益

h_{21}: 正向电压增益　h_{12} 输出导纳

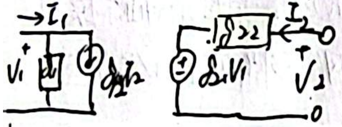


g参数: 逆~

$$\begin{bmatrix}I_1\\V_2\end{bmatrix}=\begin{bmatrix}g_{11}&g_{12}\\g_{21}&g_{22}\end{bmatrix}\begin{bmatrix}V_1\\I_2\end{bmatrix}$$

g_{11} 输入导纳　g_{12} 反向电流~

g_{21} 正向电压~　g_{22} 输出阻抗



$\Delta Z=\det[Z]$

旧量

新量

	[Z]	[Y]	[T]	[T']	[h]	[g]
[Z]	$\begin{matrix}Z_{11}&Z_{12}\\Z_{21}&Z_{22}\end{matrix}$	$\begin{matrix}\frac{Y_{22}}{\Delta_Y}&-\frac{Y_{12}}{\Delta_Y}\\-\frac{Y_{21}}{\Delta_Y}&\frac{Y_{11}}{\Delta_Y}\end{matrix}$	$\begin{matrix}\frac{A}{C}&\frac{\Delta_T}{C}\\\frac{1}{C}&\frac{D}{C}\end{matrix}$	$\begin{matrix}\frac{D'}{C'}&\frac{1}{C'}\\\frac{\Delta_{T'}}{C'}&\frac{A'}{C'}\end{matrix}$	$\begin{matrix}\frac{\Delta_h}{h_{22}}&\frac{h_{12}}{h_{22}}\\-\frac{h_{21}}{h_{22}}&\frac{1}{h_{22}}\end{matrix}$	$\begin{matrix}\frac{1}{g_{11}}&-\frac{g_{12}}{g_{11}}\\\frac{g_{21}}{g_{11}}&\frac{\Delta_g}{g_{11}}\end{matrix}$
[Y]	$\begin{matrix}\frac{Z_{22}}{\Delta_Z}&\frac{-Z_{12}}{\Delta_Z}\\-\frac{Z_{21}}{\Delta_Z}&\frac{Z_{11}}{\Delta_Z}\end{matrix}$	$\begin{matrix}Y_{11}&Y_{12}\\Y_{21}&Y_{22}\end{matrix}$	$\begin{matrix}\frac{D}{B}&-\frac{\Delta_T}{B}\\-\frac{1}{B}&\frac{A}{B}\end{matrix}$	$\begin{matrix}\frac{A'}{B'}&-\frac{1}{B'}\\-\frac{\Delta_{T'}}{B'}&\frac{D'}{B'}\end{matrix}$	$\begin{matrix}\frac{1}{h_{11}}&-\frac{h_{12}}{h_{11}}\\\frac{h_{21}}{h_{11}}&\frac{\Delta_h}{h_{11}}\end{matrix}$	$\begin{matrix}\frac{\Delta_g}{g_{22}}&\frac{g_{12}}{g_{22}}\\-\frac{g_{21}}{g_{22}}&\frac{1}{g_{22}}\end{matrix}$
[T]	$\begin{matrix}\frac{Z_{11}}{Z_{21}}&\frac{\Delta_Z}{Z_{21}}\\\frac{1}{Z_{21}}&\frac{Z_{22}}{Z_{21}}\end{matrix}$	$\begin{matrix}-\frac{Y_{22}}{Y_{21}}&-\frac{1}{Y_{21}}\\-\frac{\Delta_Y}{Y_{21}}&-\frac{Y_{11}}{Y_{21}}\end{matrix}$	$\begin{matrix}A&B\\C&D\end{matrix}$	$\begin{matrix}\frac{D'}{\Delta_{T'}}&\frac{B'}{\Delta_{T'}}\\\frac{C'}{\Delta_{T'}}&\frac{A'}{\Delta_{T'}}\end{matrix}$	$\begin{matrix}-\frac{\Delta_h}{h_{21}}&-\frac{h_{11}}{h_{21}}\\-\frac{h_{22}}{h_{21}}&-\frac{1}{h_{21}}\end{matrix}$	$\begin{matrix}\frac{1}{g_{21}}&\frac{g_{22}}{g_{21}}\\\frac{g_{11}}{g_{21}}&\frac{\Delta_g}{g_{21}}\end{matrix}$
[T']	$\begin{matrix}\frac{Z_{22}}{Z_{12}}&\frac{\Delta_Z}{Z_{12}}\\\frac{1}{Z_{12}}&\frac{Z_{11}}{Z_{12}}\end{matrix}$	$\begin{matrix}-\frac{Y_{11}}{Y_{12}}&-\frac{1}{Y_{12}}\\-\frac{\Delta_Y}{Y_{12}}&-\frac{Y_{22}}{Y_{12}}\end{matrix}$	$\begin{matrix}\frac{D}{\Delta_T}&\frac{B}{\Delta_T}\\\frac{C}{\Delta_T}&\frac{A}{\Delta_T}\end{matrix}$	$\begin{matrix}A'&B'\\C'&D'\end{matrix}$	$\begin{matrix}\frac{1}{h_{12}}&\frac{h_{11}}{h_{12}}\\\frac{h_{22}}{h_{12}}&\frac{\Delta_h}{h_{12}}\end{matrix}$	$\begin{matrix}-\frac{\Delta_g}{g_{12}}&-\frac{g_{22}}{g_{12}}\\-\frac{g_{11}}{g_{12}}&\frac{1}{g_{12}}\end{matrix}$
[h]	$\begin{matrix}\frac{\Delta_Z}{Z_{22}}&\frac{Z_{12}}{Z_{22}}\\-\frac{Z_{21}}{Z_{22}}&\frac{1}{Z_{22}}\end{matrix}$	$\begin{matrix}\frac{1}{Y_{11}}&-\frac{Y_{12}}{Y_{11}}\\\frac{Y_{21}}{Y_{11}}&\frac{\Delta_Y}{Y_{11}}\end{matrix}$	$\begin{matrix}\frac{B}{D}&\frac{\Delta_T}{D}\\-\frac{1}{D}&\frac{C}{D}\end{matrix}$	$\begin{matrix}\frac{B'}{A'}&\frac{1}{A'}\\-\frac{\Delta_{T'}}{A'}&\frac{C'}{A'}\end{matrix}$	$\begin{matrix}h_{11}&h_{12}\\h_{21}&h_{22}\end{matrix}$	$\begin{matrix}\frac{g_{22}}{\Delta_g}&\frac{g_{12}}{\Delta_z}\\-\frac{g_{21}}{\Delta_g}&\frac{g_{11}}{\Delta_g}\end{matrix}$
[g]	$\begin{matrix}\frac{1}{Z_{11}}&-\frac{Z_{12}}{Z_{11}}\\\frac{Z_{21}}{Z_{11}}&\frac{\Delta_Z}{Z_{11}}\end{matrix}$	$\begin{matrix}\frac{\Delta_Y}{Y_{22}}&\frac{Y_{12}}{Y_{22}}\\-\frac{Y_{21}}{Y_{22}}&\frac{1}{Y_{22}}\end{matrix}$	$\begin{matrix}\frac{C}{A}&-\frac{\Delta_T}{A}\\\frac{1}{A}&\frac{B}{A}\end{matrix}$	$\begin{matrix}\frac{C'}{D'}&-\frac{\Delta_T}{A}\\\frac{1}{A}&\frac{B}{A}\end{matrix}$	$\begin{matrix}\frac{h_{22}}{\Delta_h}&-\frac{h_{12}}{\Delta_h}\\-\frac{h_{21}}{\Delta_h}&\frac{h_{11}}{\Delta_h}\end{matrix}$	$\begin{matrix}g_{11}&g_{12}\\g_{21}&g_{22}\end{matrix}$

互易性: 两个端口间相互传输性质完全一致　(自由度≤3)

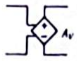
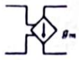
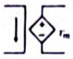
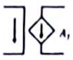
$Z_{12}=Z_{21}$, $Y_{12}=Y_{21}$, $AD-BC=1$, $h_{12}=-h_{21}$

对称性: 端口对调后一致.

$Z_{11}=Z_{22}$, $Y_{11}=Y_{22}$, $A=D$, $h_{11}h_{22}-h_{12}h_{21}=1$ 自由度≤2

有源性: 网络可对外供能 含独立源 或网络方程非齐次.

note: 方程无因果性!

		Z参数	Y参数	T参数	T'参数	h参数	g参数
压控电压源 (VCVS)	 A_v	[不存在]	[不存在]	$\begin{bmatrix}A_v^{-1}&0\\0&0\end{bmatrix}$	[不存在]	[不存在]	$\begin{bmatrix}0&0\\A_v&0\end{bmatrix}$
压控电流源 (VCCS)	 g_m	[不存在]	$\begin{bmatrix}0&0\\g_m&0\end{bmatrix}$	$\begin{bmatrix}0&-g_m^{-1}\\0&0\end{bmatrix}$	[不存在]	[不存在]	[不存在]
流控电压源 (CCVS)	 r_m	$\begin{bmatrix}0&0\\r_m&0\end{bmatrix}$	[不存在]	$\begin{bmatrix}0&0\\r_m^{-1}&0\end{bmatrix}$	[不存在]	[不存在]	[不存在]
流控电流源 (CCCS)	 A_i	[不存在]	[不存在]	$\begin{bmatrix}0&0\\0&-A_i^{-1}\end{bmatrix}$	[不存在]	$\begin{bmatrix}0&0\\A_i&0\end{bmatrix}$	[不存在]

逆因果表达

网络参数不存在的原因: ①方程不以变量为电路关系

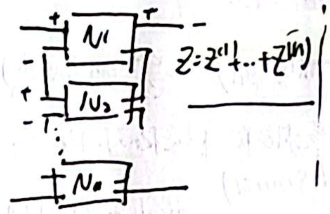
②方程中不以变量之间为控制量与被控量关系

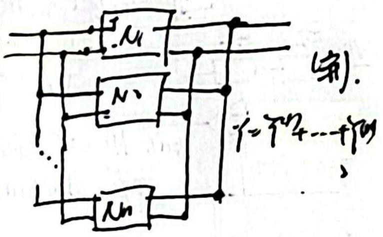
二端口网络的相互连接:

1. 级联: $[T_1][T_2]\cdots T=T^{(1)}T^{(2)}\cdots T^{(n)}$

2. 串联: 各子网络端口电流相同, 端口电压相加

(串)





(并).

3. 并联: 各子网络端口电压相同, 电流相加

note: laplace变换　$Y(s)=\mathcal{L}(y(t))=\int_{0^-}^{\infty}y(t)e^{-st}dt$

时域 $f(t)$ $(t\geq0)$	$F(s)$	性质	对应关系
$\delta(t)$	1	线性	$af+bg\leftrightarrow aF+bG$
$u(t)$ (阶跃)	$\frac{1}{s}$	微分	$f'\leftrightarrow sF-f(0^-)$
$t\,u(t)$	$\frac{1}{s^2}$	积分	$\int_0^t f\leftrightarrow\frac{F}{s}$
$e^{-at}u(t)$	$\frac{1}{s+a}$	时移	$f(t-\tau)\leftrightarrow e^{-s\tau}F$
$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$	s 移	$e^{-at}f(t)\leftrightarrow F(s+a)$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$	终值	$f(\infty)=\lim_{s\to0}sF(s)$
$e^{-at}\cos(\omega t)u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$	初值	$f(0^+)=\lim_{s\to\infty}sF(s)$

表 1: 常用拉普拉斯变换对（左）与基本性质（右）。终值定理需系统稳定才成立。

极点位置	s 域项	对应时域模态 $y(t)$
负实轴 $p=-a$ $(a>0)$	$\frac{1}{s+a}$	e^{-at}（指数衰减）
原点 $p=0$	$\frac{1}{s}$	1（阶跃/常数）
正实轴 $p=+a$	$\frac{1}{s-a}$	e^{+at}（指数发散，不稳定）
虚轴 $p=\pm j\omega_0$	$\frac{\omega_0}{s^2+\omega_0^2}$	$\sin\omega_0 t$（等幅振荡）
左半平面共轭 $-\sigma\pm j\omega_d$	$\frac{\omega_d}{(s+\sigma)^2+\omega_d^2}$	$e^{-\sigma t}\sin\omega_d t$（衰减振荡）
重极点 p（m 重）	$\frac{1}{(s-p)^k}$	$\frac{t^{k-1}}{(k-1)!}e^{pt}$（带 t 的增长因子）

$V(s)=V_s(s)\cdot H(s)$ $\xrightarrow{\text{逆换}}$ 时域

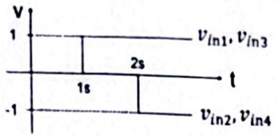
扫描全能王 创建

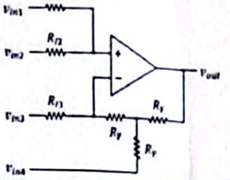
(GBW) of 8 rad/s, $R_{f1} = 2k\Omega, R_{f2} = 2k\Omega, R_{f3} = 5k\Omega, R_F = 10k\Omega$

1. In DC steady state, find the expression for v_{out} in terms of $v_{in1}, v_{in2}, v_{in3}$, and v_{in4}.

2. Find the input resistance seen from the terminals $v_{in1}, v_{in2}, v_{in3}$, and v_{in4}, respectively. (Hint: When calculating the input resistance of a specific terminal, all other voltage sources should be set to zero.)

3. If $v_{in1}, v_{in2}, v_{in3}$, and v_{in4} change according to the given timing diagram, find the time-domain signal expression for v_{out}. (Hint: what is the beta factor of this feedback circuit?)





1). $V_{out} = 4V_{in1}+4V_{in2}-6V_{in3}-V_{in4}$

2) 注意R指输入端→地的阻值

3) 先求反馈β. (定义 Vin中反Vout扰动那些部分)

将外部源置零: $\frac{0-V_-}{R_{f3}} = \frac{V_- - V_3}{R_F}$. $(\frac{V_-}{R_F}+\frac{V_-}{R_{f3}}) = \frac{V_3}{R_F}$

$\frac{V_3-V_-}{R_F}+\frac{V_3-0}{R_F}+\frac{V_3-V_{out}}{R_F}=0 \Rightarrow 8V_- = V_{out}$

$(H(s) = \frac{GBW}{s})$

$\beta = \frac{V_-}{V_{out}} = \frac{1}{8}$

闭环传函 $\frac{A(s)}{1+\beta A(s)} = \frac{H(s)}{1+H(s)\beta} = \frac{8/s}{1+8/s\cdot 1/8} = \frac{8}{s+1}$

$V_{out} = \frac{A(s)}{1+\beta A(s)} \cdot (V_{in+} - \gamma)$

$V_{in+} = \frac{V_{in1}+V_{in2}}{2}$, γ为外部驱动在负向造成影响

$\gamma + \beta V_{out} = V_{in}^- \Rightarrow \gamma = \frac{1}{8}(6V_{in3}+V_{in4})$.

代入. $V_{out}(s) = \frac{4V_{in1}+4V_{in2}-6V_{in3}+V_{in4}}{s+1} = \frac{V_{tgt}(s)}{s}$

时域: $V_{in1}=V_{in3}=u(t-1)$ $V_{in2}=V_{in4}=-u(t-1)$

$V_{in1}(s) = \frac{e^{-s}}{s}$. $V_{in2}(s)=V_{in4}(s) = -\frac{e^{-2s}}{s}$ (用 $u(t-t_0) \leftarrow \frac{e^{-t_0 s}}{s}$).

$V_{tgt}(s) = \frac{(4-6)e^{-s}}{s} + \frac{(-4+1)e^{-2s}}{s} = -\frac{2e^{-s}}{s} - \frac{3e^{-2s}}{s}$

$V_{out}(s) = -\frac{2e^{-s}}{s(s+1)} - \frac{3e^{-2s}}{s(s+1)}$ 其中 $\frac{1}{s(s+1)} = \frac{1}{s} - \frac{1}{s+1}$

$L^{-1}(\frac{1}{s} - \frac{1}{s+1}) = (1-e^{-t})u(t)$ 再由时移 $e^{-t_0 s}F(s) \leftrightarrow f(t-t_0)\cdot u(t-t_0)$

$V_{out}(t) = -2(1-e^{-(t-1)})u(t-1) - 3(1-e^{-(t-2)})u(t-2)$

Analyze the following circuits and find V_o and V_3. Assume that the op-amps are ideal and are not saturated.

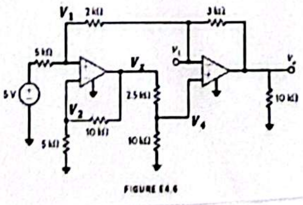


FIGURE E4.6

0.J. 本题数值有一定问题. 正反馈分析如下:

先设负反馈、由节点分析有 $\frac{V_1-5}{5K} + \frac{V_1-V_3}{2K} = 0$

$\frac{V_2-0}{5K}+\frac{V_2-V_x}{10K}=0$ $\frac{V_3-V_1}{2K}+\frac{V_3-V_0}{3K}=0$

$\frac{V_4-V_x}{2.5K}+\frac{V_4-0}{10K}=0$ $V_1=V_2$. $V_3=V_4$

则 $V_1=V_2=-2V$ $V_3=V_4=-4.8V$. $V_0=-9V$

然而, 我设 V_x 发生 e_x 扰动. 则. V_2产生扰动 $e_2 = \frac{5}{5+10}e_x = \frac{1}{3}e_x$

V_4产生扰动, $e_4 = \frac{10}{10+2.5}e_x = \frac{4}{5}e_x$

若OP2负反馈则 $e_3 = e_4 = \frac{4}{5}e_x$ 则 V_1扰动 $e_1 = \frac{5}{5+2}e_3 = \frac{4}{7}e_x$

OP1差分输入 $V_1-V_2 = (\frac{4}{7}-\frac{1}{3})e_0 > 0$.

⇒OP1正反馈

再设OP2输出扰动e_0则. V_3产生扰动 $\frac{5+2}{5+2+3}e_0 = \frac{7}{10}e_0$.

V_1扰动 $\frac{5}{5+2+3}e_0 = \frac{1}{2}e_0$.

若OP1在负反馈则 $e_1=e_2=\frac{1}{2}e_0$. V_x扰动 $= e_x\ \frac{10+5}{5}e_2 = \frac{3}{2}e_0$

V_4扰动: $e_4 = \frac{10}{10+2.5}e_x = \frac{6}{5}e_0$

$V_4-V_3 = (\frac{6}{5}-\frac{7}{10})e_0 > 0$ → OP2 工作在正反馈上

扫描全能王 创建